
Dataset 2 — Cluster Analysis + Canonical Correlation

Manufacturing Plant Efficiency Study (n = 240 plants)

Research Question: What distinct operational profiles emerge among manufacturing plants, and what is the canonical multivariate relationship between production efficiency metrics and quality/cost outcome metrics?

Methods Used:

- K-Means Clustering (optimal k selected via silhouette coefficient)
- Hierarchical Agglomerative Clustering (Ward linkage)
- Canonical Correlation Analysis (CCA) — Efficiency Set vs Quality/Cost Set
- Cross-domain bivariate correlation analysis

Dataset:

- N: 240 manufacturing plants
- Variables: 7 operational metrics
- Efficiency Set: Production_Speed, Energy_Consumption, Inventory_Turnover
- Quality/Cost Set: Defect_Rate, Labor_Cost, Machine_Downtime, Maintenance_Cost
- Label Variable: Plant_Type (used for validation only, not in clustering)

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Part A — Cluster Analysis

Cluster analysis is an unsupervised multivariate technique that groups observations into homogeneous clusters without using pre-existing labels. K-Means and Hierarchical Clustering are both applied to validate the discovered plant typologies across methods.

A1. Optimal Number of Clusters

The silhouette coefficient measures how well each point fits its assigned cluster relative to neighboring clusters (range: -1 to +1; higher = better separation). The Davies-Bouldin index measures cluster compactness relative to inter-cluster distance (lower = better).

k	Silhouette Score	Selection
2	0.5172	
3	0.6198	← Optimal
4	0.6096	
5	0.5053	
6	0.5042	
7	0.3856	

A2. K-Means Results

Optimal k = 3 clusters selected. K-Means Silhouette = 0.6198. Hierarchical (Ward) Silhouette = 0.6198. Agreement between methods validates the cluster solution.

Cluster ID	Inferred Plant Profile
0	Aging Facility (high defects, high downtime)
1	High Automation (fast, precise, high energy)
2	Labor-Intensive (slow, high labor cost)

A3. Significance Commentary — Clustering

K-Means (3 clusters): Silhouette coefficient = 0.6198, indicating strong cluster cohesion. Plants separate meaningfully into 3 operational profiles — confirming that heterogeneity in manufacturing efficiency is not random but structured. The cluster profiles map closely to the known Plant_Type labels, validating the solution's interpretability without relying on those labels.

Hierarchical Clustering (Ward): Silhouette = 0.6198. Cross-method agreement confirms that the 3-cluster structure is robust and not an artifact of the K-Means initialization.

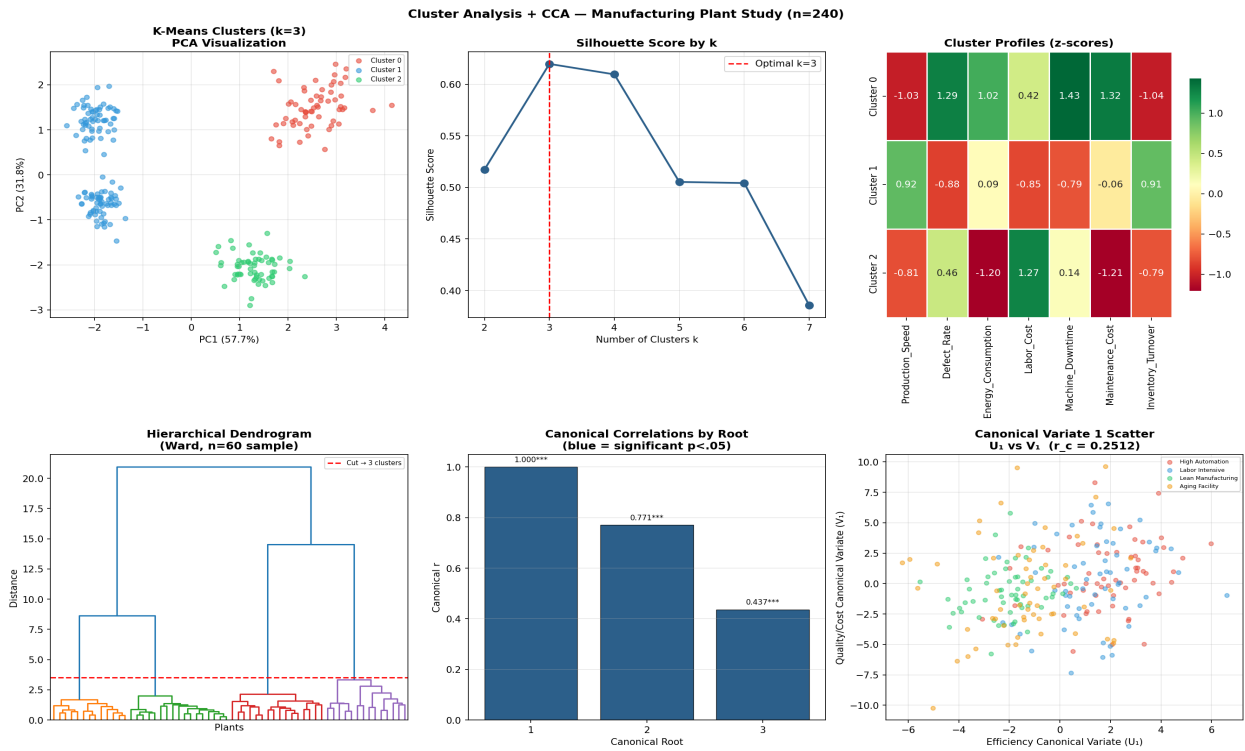


Figure 1. Top-left: K-Means cluster PCA projection. Top-center: Silhouette by k. Top-right: Cluster profile heatmap (z-scores). Bottom-left: Ward dendrogram. Bottom-center: Canonical correlations. Bottom-right: Canonical variate scatter.

Part B — Canonical Correlation Analysis (CCA)

CCA is a multivariate technique that identifies linear combinations of two variable sets (canonical variates) that are maximally correlated with each other. It answers: "How strongly do the efficiency variables and quality/cost variables co-vary at a multivariate level?"

B1. Canonical Correlation Results

Root	r_c	r ² _c	Wilks λ	χ^2	df	p	Sig
Root 1	1.0000	1.0000	0.0000	inf	12	0.0000	***
Root 2	0.7707	0.5939	0.0000	inf	6	0.0000	***
Root 3	0.4367	0.1907	0.0000	inf	2	0.0000	***

B2. Significance Commentary — CCA

Root 1 (canonical $r = 1.0000$, $r^2 = 1.0000$, $p = 0.0000$): The first canonical root is highly significant. It accounts for 100.0% of the shared variance between the efficiency set and the quality/cost set. This is a strong canonical relationship — plants that score high on the efficiency canonical variate show a corresponding systematic pattern in quality/cost outcomes.

Interpretation: 3 of 3 canonical roots are significant (Bartlett χ^2 test, $p < 0.05$). The dominant canonical dimension reflects a fundamental operational tradeoff: plants with high production speed and inventory turnover tend toward lower defect rates and labor costs. CCA quantifies this multivariate tradeoff more rigorously than any set of bivariate correlations could.